nth Roots, Radicals, and Rational Exponents

Algebra 2 Lesson 5-1 Name _____ Date _____ Period _____

Learning goal: I can relate rational exponents to roots, evaluate expressions with rational exponents, and solve equations by taking the nth root of both sides.

Language goal: I can explain what a rational exponent means, and how I took a root to solve an equation, using the words root, power, index and radicand.

Spanish: *índice* (index), *radicando* (radicand), *exponente* (exponent), *símbolo radical* (radical symbol).

On the quiz: (1) the real fourth roots of a positive number: two, ± (full Savvas quiz only; Example 5B). (2) pick every form equal to a rational exponent and (3) say what a rational exponent means, then evaluate it: Example 2, Try It 2, Practice. (4) simplify an nth root with variables (full Savvas quiz only; Example 4, IXL LQX). (5) solve $ax^3 = c$: Example 5, Try It 5. Questions 3 and 5 count 10 points each; 2 is the +5 bonus.

Example 2: Understand Rational Exponents

A. What is the meaning of the exponent in the expression $16^{\frac{1}{4}}$?

Extend the properties of integer exponents.

$$\begin{aligned} 16^{\frac{1}{4}} &= \left(2^4\right)^{\frac{1}{4}} && 16 = 2^4 \\ &= 2^{4\cdot\frac{1}{4}} = 2^1 && \text{Power of a Power} \\ &= 2 \end{aligned}$$

Since $2^4 = 16$, $16^{\frac{1}{4}}$ is $\sqrt[4]{16}$, the fourth **root** of 16, which is 2.

In general (n a positive integer), $x^{\frac{1}{n}} = \sqrt[n]{x}$, the principal nth root: one value, real when $x \geq 0$ or n is odd.

B. Write $27^{\frac{2}{3}}$ as a radical.

$$\begin{aligned} 27^{\frac{2}{3}} &= \left(27^{\frac{1}{3}}\right)^2 && \text{Power of a Power} \\ &= \left(\sqrt[3]{27}\right)^2 \\ &= 3^2 = 9 \end{aligned}$$

27 is a perfect cube (3^3), so take the cube root first, then square: $\left(\sqrt[3]{27}\right)^2 = 9$.

Same value the other way (power first, then root):
$\sqrt[3]{27^2} = \sqrt[3]{729} = 9$.

The rule. When $\sqrt[n]{c}$ is real, $c^{\frac{m}{n}} = \left(\sqrt[n]{c}\right)^m = \sqrt[n]{c^m}$.
Denominator n = the **index** (which **root**); numerator m = the **power**; c = the **radicand**.

Say it: In $27^{\frac{2}{3}}$, the denominator 3 is the _____ of the radical, so I take the cube _____. The numerator 2 is the _____, so I square. The _____ is 27.

Try It 2: Explain what each fractional exponent means, then evaluate

a. $25^{\frac{1}{2}}$

$25^{\frac{1}{2}}$ is the _____ root of 25.

_____

_____

Value: _____

b. $32^{\frac{2}{5}}$

$32^{\frac{2}{5}}$ is the _____ root of 32, _____.

_____

_____

Value: _____

Practice: Which forms are equal?

Select **all** the expressions that are equivalent to $y^{\frac{2}{5}}$. Use the rule: denominator = index, numerator = power.

☐ **A.** $\sqrt[5]{y^2}$ ☐ **B.** $\left(y^{\frac{1}{5}}\right)^2$ ☐ **C.** $\sqrt{y^5}$

☐ **D.** $y^{\frac{10}{4}}$ ☐ **E.** $(y^2)^{\frac{1}{5}}$ ☐ **F.** $\left(y^{\frac{1}{2}}\right)^{\frac{1}{5}}$

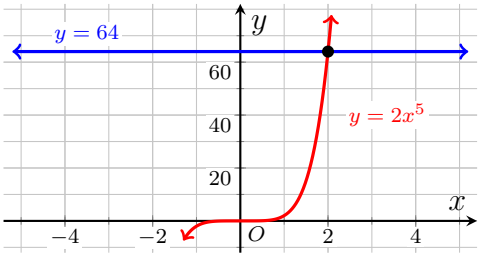
For one choice you did *not* select, write it as a single power of y and say what went wrong.

Example 5: Use nth Roots to Solve Equations

The idea: get x^n alone, then take the nth **root** of both sides: raise each side to the $\frac{1}{n}$ **power**. Odd n: $(x^n)^{\frac{1}{n}} = x$, one solution. Even n: $(x^n)^{\frac{1}{n}} = |x|$, so $x^n = d$ with $d > 0$ gives $x = \pm d^{\frac{1}{n}}$ (no real solution if $d < 0$).

A. Solve $2x^5 = 64$.

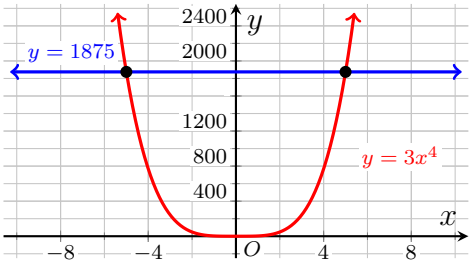
$2x^5 = 64$	
$x^5 = 32$	divide by 2
$\left(x^5\right)^{\frac{1}{5}} = 32^{\frac{1}{5}}$	raise to the $\frac{1}{5}$
$x = 2$	$2^5 = 32$



The graphs of $y = 2x^5$ and $y = 64$ meet once, so there is only one real solution: $x = 2$.

B. Solve $3x^4 = 1875$.

$3x^4 = 1875$	
$x^4 = 625$	divide by 3
$x = \pm\,(625)^{\frac{1}{4}}$	even index, $625 > 0$: $\pm$
$x = \pm 5$	$5^4 = 625$



$(-5)^4 = 625$ too: a negative number to an even power is positive. The graphs meet twice, so the solutions are $x = 5$ and $x = -5$.

$x^n = c$, **index** n **odd**	**index** n **even,** $c > 0$	**index** n **even,** $c < 0$
one real root, same sign as c: $x^3 = -8$, $x = -2$	two real roots, $\pm$: $x^4 = 625$, $x = \pm 5$	no real root: $x^4 = -256$

Say it: To solve $3x^4 = 1875$, I divided both sides by _____, then took the fourth _____ of both sides (raised them to the $\frac{1}{4}$ _____). The _____ 4 is even and the _____ 625 is positive, so there are _____ real solutions.

Try It 5: Solve each equation

a. Solve the equation $5x^3 = 320$.

$x =$ _____

How many real solutions? _____

b. Solve the equation $2p^4 = 162$.

$p =$ _____

How many real solutions? _____

Why does part b have more solutions than part a? Use the words *index* and *even*.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
9TM ★	Convert between rational exponents and radicals	Quiz 2, 3 (Example 2)
KJX ★	Evaluate rational exponents	Quiz 3 (Example 2, Try It 2)
U42 ★	Nth roots	Quiz 5 (Example 5); Quiz 1, ± (full Savvas quiz only)
EUH	Roots of integers	Warm-up, optional: the perfect powers 32, 243, 625
LQX	Simplify radical expressions with variables I	Quiz 4 (full Savvas quiz only), optional

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Extension (self-paced): Practice 50 (DOK 3, Performance Task)

A milk processing company uses cylindrical-shaped containers. The height of the container is equal to the diameter of the base. The volume of a container is 169.65 ft^3.

Part A How much material is needed to make the lateral surface area of the shipping container?

Part B The cargo hold of a ship is 20 ft high. What is the largest number of these shipping containers that could be stacked on top of each other inside the cargo hold?

Hint: write V using r only, then solve $r^3 = c$ as in Example 5.

Part A: _____ ft^2 Part B: _____ containers

Before the quiz, finish these sentences.

In $c^{\frac{m}{n}}$, the denominator n is the _____, the numerator m is the _____, and c is the _____.

To solve $ax^n = c$, I divide by a, then take the nth _____ of both sides. If the index is even and $\frac{c}{a} > 0$, I write _____ (two real solutions); if the index is even and $\frac{c}{a} < 0$, there is _____ real solution.